Project Paris-Duchesse: Mathematical Formulation

M1 IDD - Mathematics of Data Science

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1. Linear Programming Relaxation

Question 1: Linear Program Formulation

We define the following sets and indices:

- $\mathcal{O} = \{A1, A2, B1, B2, B3, C1, C2, D1, D2, D3, E1, E2\}$ is the set of 12 offices.
- $\mathcal{P} = \{0, 1, 2, 3, 4, 5\}$ is the set of phases.
- $\mathcal{K} = \{P1..P4, S1..S4, T1..T4, M1..M4, O1..O2\}$ is the set of 18 occupants.
- $\mathcal{R}_p \subset \mathcal{O}$ is the set of offices under renovation during phase p.

Let $x^k_{i,j,p} \in [0, 1]$ be the decision variable representing the fraction of occupant k moving from office i to office j at the start of phase p.

Primal Problem (P)

$$
\begin{aligned}
\underset{x}{\text{minimize}} \quad & \sum_{p=1}^{5} \sum_{k\in\mathcal{K}} \sum_{i\in\mathcal{O}} \sum_{j\in\mathcal{O}} c_{ij} x^k_{i,j,p} \\
\text{subject to} \quad & \sum_{j\in\mathcal{O}} x^k_{s_k,j,1} = 1, \quad \forall k \in \mathcal{K} \\
& \sum_{i\in\mathcal{O}} x^k_{i,j,p} - \sum_{l\in\mathcal{O}} x^k_{j,l,p+1} = 0, \quad \forall k \in \mathcal{K}, \forall j \in \mathcal{O}, \forall p \in \{1, \ldots, 4\} \\
& \sum_{k\in\mathcal{K}} \sum_{i\in\mathcal{O}} x^k_{i,j,p} \leq 2, \quad \forall j \in \mathcal{O}, \forall p \in \mathcal{P} \\
& x^k_{i,j,p} = 0, \quad \forall k, \forall i, \text{if } j \in \mathcal{R}_p \\
& \sum_{i\in\mathcal{O}} x^k_{i,j,5} = \delta_{j,\text{target}(k)}, \quad \forall k \in \mathcal{K}, \forall j \in \mathcal{O} \\
& x^k_{i,j,p} \geq 0.
\end{aligned}
\tag{1}
$$

where $c_{ij} = 1$ if $i \neq j$ and 0 otherwise.

KKT Conditions

The constraints are linear, satisfying constraint qualification. The Lagrangian is:

$$
\mathcal{L}(x, \lambda, \mu, \nu) = \sum c_{ij} x + \sum \lambda^{flow}_{k,j,p} \left(\sum x_{in} - \sum x_{out}\right) + \sum \mu^{cap}_{j,p} \left(\sum x - 2\right) - \sum \nu^k_{i,j,p} x \tag{2}
$$

The necessary and sufficient conditions for optimality are:

$$\frac{\partial \mathcal{L}}{\partial x_{i,j,p}^k} = c_{ij} + \lambda_{j,p,k}^{flow} - \lambda_{i,p-1,k}^{flow} + \mu_{j,p}^{cap} - \nu_{i,j,p}^k = 0 \tag{3}$$

$$\sum_i x_{i,j,p}^k - \sum_l x_{j,l,p+1}^k = 0 \tag{4}$$

$$\mu_{j,p}^{cap} \geq 0, \quad \sum_{k,i} x_{i,j,p}^k - 2 \leq 0, \quad \mu_{j,p}^{cap}\left(\sum_{k,i} x_{i,j,p}^k - 2\right) = 0 \tag{5}$$

$$\nu_{i,j,p}^k \geq 0, \quad x_{i,j,p}^k \geq 0, \quad \nu_{i,j,p}^k x_{i,j,p}^k = 0 \tag{6}$$

Question 3: Separating Presidency and Students

We define the adjacency graph $G = (\mathcal{O}, E)$ where $E = \{(A1, A2), (A2, B1), (A2, D3), \dots\}$. Let $\mathcal{K}_P$ be the set of Presidency occupants and $\mathcal{K}_S$ be the set of Students. We add the following exclusion constraints for all phases p and all adjacent pairs $(u, v) \in E$:

$$\left(\sum_{k \in \mathcal{K}_P} \sum_{i \in \mathcal{O}} x_{i,u,p}^k\right) + \left(\sum_{k \in \mathcal{K}_S} \sum_{j \in \mathcal{O}} x_{j,v,p}^k\right) \leq 1 \tag{7}$$

This constraint ensures that if any Presidency member is in office u, no Student can be in the adjacent office v.

2. Penalized Problem

Question 4:

We relax the hard constraint on the final position and instead penalize the distance to the target configuration $N^F \in \mathbb{R}^{|\mathcal{O}| \times |\mathcal{S}|}$ (where $N_{j,S}^F$ is the target count of service S in office j) starting from phase 0. Let $y_{j,p,S} = \sum_{k \in S} \sum_i x_{i,j,p}^k$ be the count of service S in office j at phase p.

Primal Problem (Penalized)

$$\min_{x,\delta} \quad \sum c^T x + \lambda \sum_{p=0}^{4} \sum_{j \in \mathcal{O}} \sum_{S \in \text{Services}} \delta_{j,p,S} \tag{8}$$

subject to the constraints of (P) (excluding the hard target constraint) plus:

$$y_{j,p,S} - N_{j,S}^F \leq \delta_{j,p,S} \tag{9}$$

$$-(y_{j,p,S} - N_{j,S}^F) \leq \delta_{j,p,S} \tag{10}$$

Dual Problem

Let $\alpha_{j,p,S} \geq 0$ and $\beta_{j,p,S} \geq 0$ be the dual variables for the two penalty inequalities. The dual involves maximizing the lower bounds subject to:

$$\alpha_{j,p,S} + \beta_{j,p,S} \leq \lambda \quad \forall j, p, S \tag{11}$$

3. Semidefinite Programming Relaxation

Question 6: SDP Relaxation

We map the binary variables $x^k_{i,j,p} \in \{0,1\}$ to $z \in \{-1,1\}$ via $z = 2x - 1$. Let $\mathbf{z}$ be the vector of all such variables. We introduce the matrix variable $Z \in \mathbb{S}^{N+1}$ to relax the rank-1 constraint of $\tilde{z}\tilde{z}^T$ where $\tilde{z} = (1, \mathbf{z})$.

To formulate the SDP, the variables are defined as follows:

- **Cost Matrix C:** A sparse matrix where the diagonal and the first row/column are populated with move costs c_{ij} to represent the linear objective in terms of Z.
- **Matrices A_m:** These are indicator matrices. Each A_m corresponds to one specific constraint (flow, capacity, or renovation). It selects the relevant indices in Z to perform the summation required by that constraint.
- **Vector b_m:** The target values for the constraints, adjusted for the z mapping. For capacity, $b_m = 4 -$ (number of variables in sum) to reflect the $\sum x \leq 2$ limit. For flow conservation, $b_m = 0$.

Primal SDP

$$\begin{array}{ll} \underset{Z}{\text{minimize}} & \text{trace}(CZ) \\ \text{subject to} & \text{trace}(A_m Z) = b_m, \quad m = 1 \dots M \\ & Z_{ii} = 1, \quad \forall i \\ & Z \succeq 0 \end{array} \tag{12}$$

Dual SDP

$$\begin{array}{ll} \underset{y,w,S}{\text{maximize}} & b^T y + \mathbf{1}^T w \\ \text{subject to} & \sum_m y_m A_m + \text{diag}(w) + S = C \\ & S \succeq 0 \end{array} \tag{13}$$

KKT Conditions

$$\text{trace}(A_m Z^*) = b_m, \quad Z^*_{ii} = 1, \quad Z^* \succeq 0 \tag{14}$$

$$\sum_m y^*_m A_m + \text{diag}(w^*) + S^* = C, \quad S^* \succeq 0 \tag{15}$$

$$Z^* S^* = 0 \tag{16}$$